

- 1 Baker's yeast, *Saccharomyces cerevisiae*, has two metabolic pathways for the production of ATP, aerobic respiration and fermentation. Oxygen is **not** used in fermentation.

These pathways are summarised in Fig. 1.1.

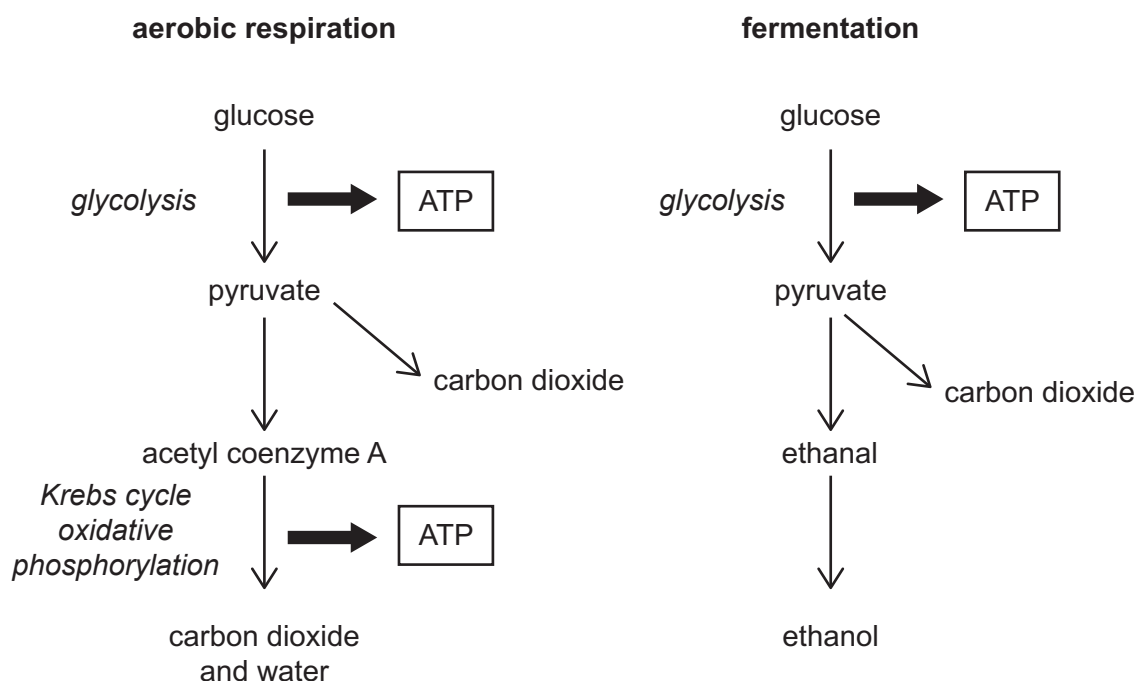


Fig. 1.1

Some students were researching details of the respiration of yeast. In their research, the students found information about the Crabtree effect. This effect occurs when yeast is kept in high concentrations of glucose. The yeast gains sufficient ATP from substrate-linked phosphorylation during glycolysis. Yeast does not carry out the Krebs cycle or oxidative phosphorylation and therefore does not use oxygen.

The students wanted to investigate the effect of temperature on the activity of yeast and decided to use yeast solutions in which the yeast obtained all of its ATP by fermentation.

- The students made a suspension of yeast from 1 g of dried yeast and 25 cm³ of water.
- The suspension was left for 2 hours at 20 °C.
- After 2 hours, 25 cm³ of glucose solution was added to the yeast suspension and the mixture was stirred.
- Some of the mixture was added immediately to the apparatus shown in Fig. 1.2.

The students measured the volume of carbon dioxide produced by yeast fermentation.

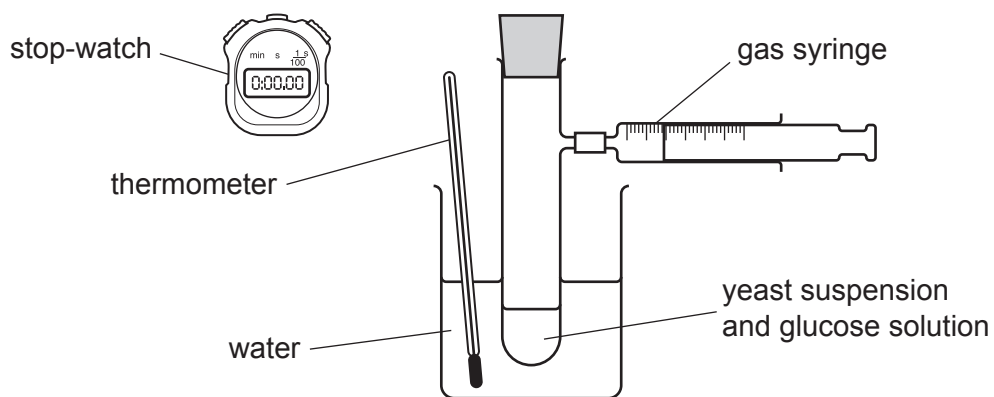


Fig. 1.2

- (a) (i) State the independent variable **and** the dependent variable in this investigation.

independent variable

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dependent variable

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[2]

- (ii) Explain why the students stirred the mixture of yeast and glucose before adding it to the tube in the apparatus in Fig. 1.2.

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..... [1]

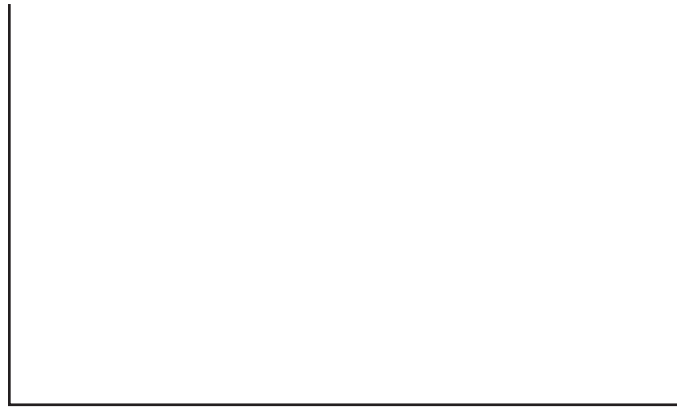
- (iii) Describe a method using the apparatus, set up as shown in Fig. 1.2, that the students could use to find the temperature at which yeast carries out fermentation at its maximum rate.

Your method should be set out in a logical order and be detailed enough for another person to follow.

You should **not** include details of how to make the yeast suspension or how to set up the apparatus.

..... [7]

- (iv) Complete the sketch graph to predict the results that you would expect from the method that you have given in part (iii). Include axes labels with units in your answer.



[2]

- (b) Scientists have discovered that some other species of yeast show the Crabtree effect.

The Crabtree effect means that in high concentrations of glucose **and** with oxygen present, pyruvate is not metabolised in mitochondria so aerobic respiration does not occur.

Yeast species that are able to maintain their levels of ATP production when conditions suddenly change and **no** oxygen is available show the Crabtree effect.

An investigation compared *S. cerevisiae* with the yeast *Candida tropicalis* to see if this other species also shows the Crabtree effect.

Three flasks containing suspensions of *S. cerevisiae* in a growth medium were initially kept in different conditions (conditions for growth).

After a set time, the yeast in each flask was transferred to a flask containing a fresh medium with high glucose concentration and **no** oxygen (experimental conditions). All other conditions were standardised.

The same procedure was repeated for *C. tropicalis*.

The conditions for growth and the experimental conditions in the flasks are summarised in Table 1.1.

Table 1.1

flask	conditions for growth	experimental conditions
flask 1	high glucose concentration with no oxygen	high glucose concentration with no oxygen
flask 2	high glucose concentration with oxygen	high glucose concentration with no oxygen
flask 3	water (no glucose) with oxygen	high glucose concentration with no oxygen

Samples taken from the flasks kept in the experimental conditions were put into the apparatus shown in Fig. 1.2 to measure the volume of carbon dioxide produced by the two yeast species.

The results are shown in Fig. 1.3.

The final volume of carbon dioxide produced by the sample of yeast from flask 1 at 30 minutes is shown as a maximum volume of 100% in Fig. 1.3. All other volumes were calculated as a proportion of this maximum. This allows a direct comparison to be made between the two species.

